IP Addressing & Subnetting

1. IPv4 Addressing

An IPv4 address is a 32-bit number that uniquely identifies a network interface on a machine. It is typically represented in dot-decimal notation (e.g., 192.168.1.1).

2. IPv4 Classes

- Class A: 1.0.0.0 to 126.0.0.0 (Large networks)
- Class B: 128.0.0.0 to 191.255.0.0 (Medium networks)
- Class C: 192.0.0.0 to 223.255.255.0 (Small networks)
- Class D: 224.0.0.0 to 239.255.255.255 (Multicast)
- Class E: 240.0.0.0 to 255.255.255.255 (Experimental)

3. Subnetting and CIDR

Subnetting is the practice of dividing a network into two or more smaller networks. A subnet mask distinguishes the network prefix from the host identifier.

CIDR (Classless Inter-Domain Routing) notation represents an IP address and its routing prefix (e.g., 192.168.1.0/24).

4. IPv6 Basics

IPv6 addresses are 128-bit identifiers, solving the IPv4 address exhaustion problem. They are represented as eight groups of four hexadecimal digits (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

5. Private vs Public IPs and NAT

Private IP addresses are non-routable on the internet (e.g., 192.168.x.x, 10.x.x.x). Public IP addresses are globally unique and routable.

NAT (Network Address Translation) maps private IP addresses to a public IP address before transferring the information to the internet, allowing multiple devices on a local network to share a single public IP.